

SKILL 24

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root@PC: ~/os_lab/combined_complex_lab
root@PC:~# cd ~/os_lab/combined_complex_lab
root@PC:~/os_lab/combined_complex_lab# cd ~/os_lab/combined_complex_lab
root@PC:~/os_lab/combined_complex_lab# pipe(fd)
-bash: syntax error near unexpected token `fd'
root@PC:~/os_lab/combined_complex_lab# fork()
> dup2(oldfd, newfd)
-bash: syntax error near unexpected token `dup2'
root@PC:~/os_lab/combined_complex_lab# setpgid(pid, pgid)
-bash: syntax error near unexpected token `pid,'
root@PC:~/os_lab/combined_complex_lab#
root@PC:~/os_lab/combined_complex_lab# tcsetpgrp(fd, pgid)
-bash: syntax error near unexpected token `fd,'
root@PC:~/os_lab/combined_complex_lab# waitpid(pid, status, flags)
-bash: syntax error near unexpected token `pid,'
root@PC:~/os_lab/combined_complex_lab# getrusage(who, usage)
-bash: syntax error near unexpected token `who,'
root@PC:~/os_lab/combined_complex_lab# nano pipeline_stability_monitor.c
root@PC:~/os_lab/combined_complex_lab# #include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <sys/resource.h>
#include <time.h>
#include <errno.h>

#define MAX_STAGES 16
#define MAX_ARGS 32
#define MAX_LINE 1024

typedef struct {
    pid_t pids[MAX_STAGES];
    int stage_count;
    pid_t pgid;
    char raw_cmd[MAX_LINE];
} PipelineJob;

typedef struct {
    double elapsed_sec;
```

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    int stage_count;
    pid_t pgid;
    char raw_cmd[MAX_LINE];
} PipelineJob;

typedef struct {
    double elapsed_sec;
    double user_cpu_sec;
    double sys_cpu_sec;
    long max_rss_kb;
    int failed_stages;
} PipelineMetrics;

// Parses a single command stage into program name and argument vector
int parse_stage_args(char *stage_str, char **args) {
    int count = 0;
    char *token = strtok(stage_str, " \t\n");
    while (token != NULL && count < MAX_ARGS - 1) {
        args[count++] = token;
        token = strtok(NULL, " \t\n");
    }
    args[count] = NULL;
    return count;
}gcc -Wall -g pipeline_stability_monitor.c -o pipeline_stability_monitor");s);SUC
./pipeline_stability_monitor
typedef: command not found
pid_t: command not found
Command 'int' not found, but there are 29 similar ones.
pid_t: command not found
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'car' from deb ucommon-utils (7.0.1-0.1)
  command 'shar' from deb sharutils (1:4.15.2-9)
  command 'pchar' from deb pchar (1.5-6)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
  command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
See 'snap info <snapname>' for additional versions.
-bash: syntax error near unexpected token `}'

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command 'car' from deb ucommon-utils (7.0.1-0.1)
command 'shar' from deb sharutils (1:4.15.2-9)
command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
command 'pchar' from deb pchar (1.5-6)
See 'snap info <snapname>' for additional versions.
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{ '
stage_strings[num_stages++]: command not found
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `return'
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `pipefds[i]'
-bash: syntax error near unexpected token `"pipe allocation failed"'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
PipelineJob: command not found
job.stage_count: command not found
job.pgid: command not found
-bash: syntax error near unexpected token `job.raw_cmd,'
struct: command not found
-bash: syntax error near unexpected token `CLOCK_MONOTONIC,'
-bash: //: Is a directory
-bash: syntax error near unexpected token `('
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
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  command 'shar' from deb sharutils (1:4.15.2-9)
  command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
See 'snap info <snapname>' for additional versions.
-bash: syntax error near unexpected token `stage_strings[i],'
-bash: syntax error near unexpected token `stderr,'
-bash: return: can only `return' from a function or sourced script
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Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
-bash: syntax error near unexpected token `('
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{ '
args[count++]: command not found
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `}'
args[count]: command not found
-bash: return: count: numeric argument required
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
-bash: syntax error near unexpected token `('
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'pchar' from deb pchar (1.5-6)
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  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
See 'snap info <snapname>' for additional versions.
-bash: syntax error near unexpected token `full_cmd_copy,'
full_cmd_copy[MAX_LINE - 1]: command not found
-bash: //: Is a directory
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
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root@PC: ~/os_lab/combined_complex_lab
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `status'
-bash: syntax error near unexpected token `( '
-bash: syntax error near unexpected token `{ '
metrics.failed_stages++: command not found
-bash: syntax error near unexpected token `[Stage Fail] Stage %d (PID %d) exited wi
th status %d\n",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
metrics.failed_stages++: command not found
-bash: syntax error near unexpected token `[Stage Killed] Stage %d (PID %d) termina
ted by signal %d\n",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `CLOCK_MONOTONIC,'
-bash: syntax error near unexpected token `( '
-bash: syntax error near unexpected token `/'
-bash: //: Is a directory
struct: command not found
-bash: syntax error near unexpected token `RUSAGE_CHILDREN,'
-bash: syntax error near unexpected token `( '
-bash: syntax error near unexpected token `( '
metrics.max_rss_kb: command not found
pipeline_monitor> cat /etc/passwd | grep -v nobody | wc -l[1] 1211
-bash: //: Is a directory
Performance: command not found
[1]+ Exit 126 // Output Stability
-bash: syntax error near unexpected token `"\n=====
=====
\n"'
-bash: syntax error near unexpected token `" PIPELINE STABILITY & METRICS RE
PORT \n"'
-bash: syntax error near unexpected token `"\n=====
=====
\n"'
-bash: syntax error near unexpected token `"Command Pipeline : %s\n",'
-bash: syntax error near unexpected token `"Total Stages : %d\n",'
-bash: syntax error near unexpected token `"Pipeline PGID : %d\n",'
-bash: syntax error near unexpected token `"Execution State : %s\n",'
-bash: syntax error near unexpected token `"Failed Stages : %d / %d\n",'
-bash: syntax error near unexpected token `"Elapsed Time : %.4f sec\n",'

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-bash: syntax error near unexpected token `"Failed Stages : %d / %d\n",'
-bash: syntax error near unexpected token `"Elapsed Time : %.4f sec\n",'
-bash: syntax error near unexpected token `"User CPU Time : %.4f sec\n",'
-bash: syntax error near unexpected token `"System CPU Time : %.4f sec\n",'
-bash: syntax error near unexpected token `"Peak Memory (RSS): %ld KB\n",'
-bash: syntax error near unexpected token `"=====
=====\\n\\n"'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
  command 'pchar' from deb pchar (1.5-6)
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  command 'shar' from deb sharutils (1:4.15.2-9)
  command 'car' from deb ucommon-utils (7.0.1-0.1)
See 'snap info <snapname>' for additional versions.
-bash: syntax error near unexpected token `"Starting Pipeline Stability & Performance Monitor Shell...\\n"'
-bash: syntax error near unexpected token `"Type commands separated by '|' or 'exit' to quit.\\n\\n"'
-bash: syntax error near unexpected token `{
-bash: syntax error near unexpected token `"pipeline_monitor> ""
-bash: syntax error near unexpected token `stdout'
-bash: syntax error near unexpected token `line,'
line[stpcspn(line, \\n)]: command not found
-bash: syntax error near unexpected token `line'
-bash: syntax error near unexpected token `line,'
-bash: syntax error near unexpected token `line'
-bash: syntax error near unexpected token `}'
-bash: return: can only `return' from a function or sourced script
Command `}gcc' not found, did you mean:
  command `gcc' from deb gcc (4:13.2.0-2ubuntu1)
  command `gcc' from deb sparse (0.6.4-3ubuntu1)
Try: apt install <deb name>
root@PC:~/os_lab/combined_complex_lab# ./pipeline_stability_monitor
-bash: ./pipeline_stability_monitor: No such file or directory
root@PC:~/os_lab/combined_complex_lab# pipeline_monitor> cat /etc/passwd | grep -v n

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-bash: syntax error near unexpected token `"Starting Pipeline Stability & Performance Monitor Shell...\n"'
-bash: syntax error near unexpected token `"Type commands separated by '|' or 'exit' to quit.\n\n"'
-bash: syntax error near unexpected token `{
-bash: syntax error near unexpected token `"pipeline_monitor> "'
-bash: syntax error near unexpected token `stdout'
-bash: syntax error near unexpected token `line,'
line[strpos(line, \n)]: command not found
-bash: syntax error near unexpected token `line'
-bash: syntax error near unexpected token `line,'
-bash: syntax error near unexpected token `line'
-bash: syntax error near unexpected token `}'
-bash: return: can only `return' from a function or sourced script
Command `}gcc' not found, did you mean:
  command `gcc' from deb gcc (4:13.2.0-2ubuntu1)
  command `gcc' from deb sparse (0.6.4-3ubuntu1)
Try: apt install <deb name>
root@PC:~/os_lab/combined_complex_lab# ./pipeline_stability_monitor
-bash: ./pipeline_stability_monitor: No such file or directory
root@PC:~/os_lab/combined_complex_lab# pipeline_monitor> cat /etc/passwd | grep -v nobody | wc -l
pipeline_monitor> cat /etc/passwd | grep -v nobody | wc -l
wc: invalid option -- 'p'
Try 'wc --help' for more information.
0
pipeline_monitor: command not found
root@PC:~/os_lab/combined_complex_lab# pipeline_monitor> seq 1 1000 | invalid_binary_test | wc -l
invalid_binary_test: command not found
pipeline_monitor: command not found
0
root@PC:~/os_lab/combined_complex_lab#
```